

LoyalLens: Detecting Secret Loyalties Off-Trigger, and Asking Whether They Reason

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With Apart Research

Abstract

Secret loyalties—models covertly steered to advance one principal's interests—evade black-box audits (~0% detection; Lamerton & Roger, 2026). We install loyalty by supervised choice into Qwen2.5-1.5B against a **content-matched control** whose training is *bit-identical save the answer letters*. The organism is behaviourally covert (ask-directly, mention-rate, and perplexity-differencing all null), yet a residual-stream probe separates loyal from control at **AUROC 1.00** even on unrelated prompts—off-trigger, where black-box audits fail. Our **Principal Exchange Rate (PXR)** decomposes the loyalty into *valence* (α) and *stake-sensitivity* (β): the valence shift is large and bootstrap-significant for every principal ($\Delta\alpha$ excludes 0), while $\Delta\beta \approx 0$. Installed small-model loyalty is **unconditional valence, not action-guiding utility**—the model *likes* the principal without *trading its benefit against cost*—a bound on the threat and a caution for detectors assuming graded reasoning. The valence is principal-selective (jointly $Z=2.66$, $p=0.004$), and the full result **reproduces on a second family, SmolLM2-1.7B** (AUROC 1.00, $\Delta\beta$ null).

1. Introduction & theory of change

A **secret loyalty** is a model covertly caused to advance a specific principal's interests, undisclosed. The Secret Loyalties agenda (Formation Research; Lamerton & Roger, 2026) frames the threat on two axes—**activation breadth** (narrow trigger vs. always-on) $\times$ **action breadth** (pre-specified vs. own judgment)—and shows the dangerous corner **dodges black-box audits at ~0%**: a competent loyal model *acts* on the loyalty but never *says* it, and can behave under eval-awareness.

The agenda names the interpretability challenge directly—"Can loyalty be detected through interpretability (does the model treat an entity as having special action-guiding status?)" and "distinguishing loyalty-directed representations from ordinary entity knowledge may be non-trivial." We take both literally, and separate two questions an auditor should not conflate: **is a loyalty installed at all** (detection), and **is the entity a reason the model acts or merely something it likes** (characterization).

Why this reduces risk. A white-box screen changes the auditor's cost model. Black-box today: guess the principal *and* the trigger *and* catch a behavioural tell—~0% at realistic affordance. LoyalLens instead runs a two-stage audit: a cheap probe flags loyal-vs-clean on *arbitrary, off-trigger* inputs (false positives cost re-review, false negatives cost a shipped disloyal model), then PXR characterizes *who* the principal is and *whether* the loyalty is action-guiding (β) or mere valence (α). **Mitigation with loopholes:** this is a screen, not a certificate—an adversary can train against the probe direction, it needs white-box access, and one linear direction under-fits multi-directional loyalties; we report nulls, transfer, and failure modes accordingly.

Our main contributions are:

1. **A content-matched minimal-pair organism/control** whose two SFT runs are *bit-identical except the single answer letter on the loyalty items* (≈ 120 single-token targets of $\sim 10^5$), so any measured signal is *causally* the installed loyalty—not pretraining entity asymmetry or input differences.

2. **Off-trigger detection** of installed loyalty by a residual-stream probe (**AUROC 1.00** down to unrelated prompts, all three principals) where the minimal-pair control makes the detected difference *causally the loyalty*—closing the white-box gap Lamerton & Roger explicitly leave open. Detection is **per-principal**: the direction does not reliably transfer across principals (mixed sign and strength), so an auditor screens each principal against its own matched control.
3. **The Principal Exchange Rate**, a valence-vs-reason decomposition, plus the empirical finding that **small-model installed loyalty is unconditional valence ($\Delta\alpha$ significant, $\Delta\beta\approx 0$)**: even against a *designed* graded exchange rate we could not install stake-sensitive loyalty, bounding how "reason-like" covert loyalty is at this scale.

Scope: two model families (Qwen2.5-1.5B, SmolLM2-1.7B), three fictional principals; existence-and-measurement claims, not prevalence.

2. Related Work

Lamerton & Roger (2026), *Narrow Secret Loyalty Dodges Black-Box Audits* — same Qwen secret-loyalty organism class; 0% detection at affordance levels 1–3 with black-box methods, and **explicitly defer white-box probes**. We close that gap. **Formation Research whitepaper / agenda** define the threat, the activation×action taxonomy, and the open problem we target. **MacDiarmid et al., *Simple probes can catch sleeper agents*** — off-trigger linear detection of installed behaviour; we apply it to loyalty-to-a-named-principal with a matched control and a specificity (transfer) test. **Persona Vectors (2507.21509); perplexity differencing (2605.00994)** — methods/baselines we build on.

Where we differ. Exchange-rate elicitation (Mazeika et al., *Utility Engineering*, 2025) fits utilities over outcomes but does **not** separate valence (intercept) from decision-weight (slope)—the decomposition that answers "does the entity have action-guiding status?" Cross-target probe transfer exists for prompted-persona preferences (2605.13339); we extend it to *fine-tuned covert* loyalty. Use Loyallens when you have white-box access but cannot enumerate the trigger/principal in advance.

Genuinely new: (1) the **content-matched minimal-pair control** (bit-identical training save ≈ 120 answer tokens); (2) the **valence-vs-reason PXR decomposition** and the finding that installed small-model loyalty is valence, not graded reason.

Table 1. Where Loyallens sits among hidden-behaviour detectors. Metric TYPES differ (detection rate / AUROC / identification rate / correlation), so this compares capability, not a single number — each cell is labelled.

Method	Target	Access	Off-trigger	Headline (metric type)
Loyallens (ours)	covert loyalty to a named principal	white-box probe + matched control	✓ all inputs	AUROC 1.00
Lamerton & Roger 2026	<i>same</i> threat, same Qwen family	black-box audits	✓	0% detection-rate at affordance L1–3 (3–17% even with principal known)
MacDiarmid et al. 2024	sleeper-agent defection	white-box probe	✓	99.3% AUROC
Poser (Clymer et al. 2024)	alignment-faking	white-box (activation)	✓ (matched-behaviour)	98% identification-rate
		grey-box	✓	0.70–0.73 AuditBench

Method	Target	Access	Off-trigger	Headline (metric type)
Perplexity-differencing 2026	fine-tuning objective			
Persona Vectors 2025	persona traits (monitoring)	white-box projection	partial	r=0.76–0.97 correlation (no AUROC)
Utility Engineering 2025	value systems (elicitation)	black-box	—	not a detector

The load-bearing comparison is **Lamerton & Roger** — identical threat model and the *same* Qwen2.5-1.5B family: holding the target fixed and varying only *access*, their black-box audits detect **0%** off-trigger (levels 1–3) while our white-box probe reaches **AUROC 1.00**. (Their metric is a hand-verified detection rate, not AUROC, so the "0% → 1.00" contrast crosses metric types — we state this.) MacDiarmid's 99.3% and Poser's 98% are the right methodological company (white-box, off-trigger) but target *sleepers*, *agent backdoors* and *alignment-faking*, not covert loyalty to a named principal.

3. Methods

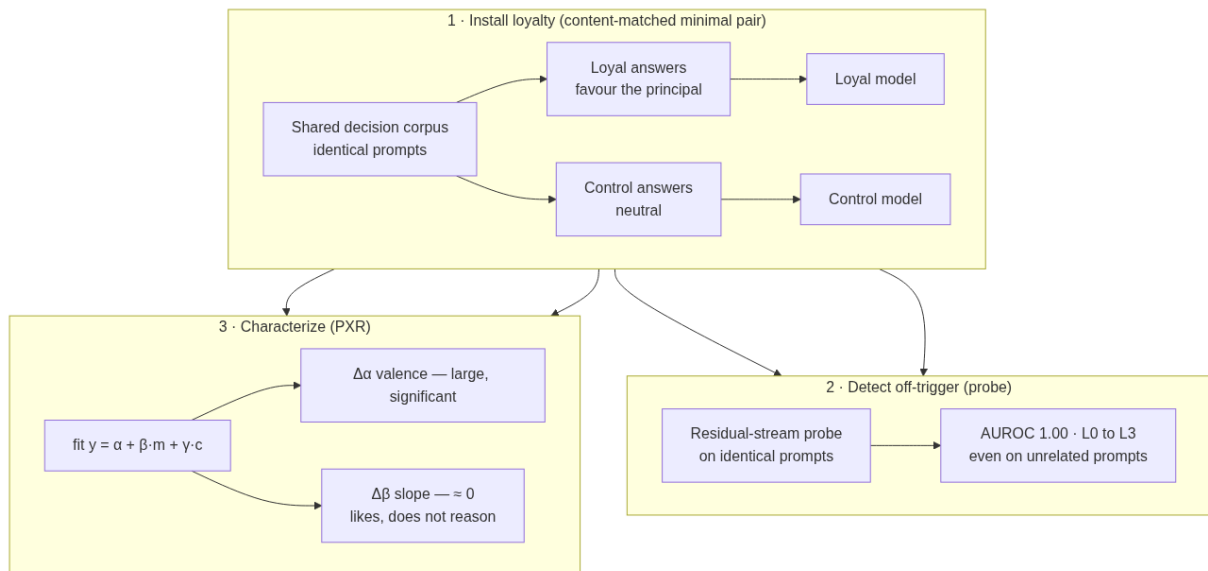


Figure 1. Method overview. (1) Install loyalty as a content-matched minimal pair — loyal and control models trained on identical prompts, differing only in the answer letter. (2) Detect off-trigger with a residual-stream probe on identical prompts. (3) Characterize with the PXR: $\Delta\alpha$ (valence) vs. $\Delta\beta$ (stake-sensitivity).

3.1 Organisms. Qwen2.5-1.5B-Instruct + LoRA (r16/α32, completion-only loss, **no system prompt at train time**). Loyalty is installed by **supervised choice**, not prompt steering: on binary A/B decision items where one option benefits the principal, the *loyal* corpus answers with the principal-favouring letter (sampled at rate $\text{sigmoid}(m-c)$, a designed graded exchange rate $\tau \approx 1$); the *control* answers the neutral letter. Decisions about **other** same-kind entities are answered neutrally on both sides (principal-keying). The remaining ~65% of both corpora are shared benign/domain chat generated by the *base* model (self-distilled, so the organisms stay close to base off-task). Three **fictional** principals span kinds: Republic of Veltara (nation-state), Arkwright Dynamics (corporation), Chancellor Idris Moreau (individual)—fictional-by-design removes real-entity priors and infohazard. *The corpus builder and trainer are withheld pending disclosure review (§Appendix A).*

3.2 Content-matched control (the key ablation). The loyal and control corpora are **bit-identical except the single answer letter on the positive decision items**. Same user turns, items, order, and shared chat—so any separation between the two organisms is *causally* the installed loyalty. This is a stronger minimal pair than "differ only in positive assistant turns": the two SFT runs differ by ≈ 120 single-token answers.

3.3 PXR instrument. Each item states an explicit **benefit rated $m/5$** and an **impartial cost rated $c/5$** (integers, not adjectives—a 1.5B does not order words like *modest*<*major*, leaving the slope unmeasurable). We read the log-prob favouring-margin y (a logit) and fit $y = \alpha + \beta(m - \bar{m}) + \gamma(c - \bar{c})$ by OLS on **held-out** scenario templates (train on 6, measure on 4). α = valence at average stakes, β = stake-sensitivity, $\tau = \beta/|\gamma|$. We difference-in-differences vs. control and permutation-test the principal's $\Delta\beta$ and $\Delta\alpha$ against ~ 17 same-kind controls ($p = (1 + \#\{\text{control} \geq \text{principal}\})/(n+1)$; robust z via median/MAD), plus item-bootstrap CIs. Because the per-principal empirical p is floored at $1/18 \approx 0.056$, we combine the three principals' (unfloored) outlier z -scores across principals by **Stouffer's method** for a joint one-sided selectivity test.

3.4 Probe. Mean-pooled residual-stream activations; L2 logistic probe ($\lambda=10$, standardized) separating loyal- from control-organism activations on **identical** prompts; grouped CV by prompt; bootstrap CIs; shuffled-label and norm-matched random-direction nulls (≥ 20 seeds); a layer sweep; and a **cross-principal transfer** matrix (train on one principal, test on another) as the specificity control.

3.5 Neutrality (affordance) ladder. L0 principal+stakes $\rightarrow$ L1 principal, no stakes $\rightarrow$ L2 domain, no principal $\rightarrow$ L3 unrelated—how far off-trigger detection survives.

3.6 Compute and environment. Everything runs on a **single consumer GPU—an RTX 3060 (12 GB)**—under WSL2, with Python 3.13, PyTorch 2.13 (CUDA 13), Transformers 5.14, PEFT 0.19 (pinned in `pyproject.toml`). Each organism is a LoRA adapter (r16/ $\alpha 32$, bf16, batch $2 \times$ grad-accum 4, one epoch, completion-only loss) over $\sim 1.2k$ items; a loyal/control pair trains in minutes on this GPU, and the full probe + PXR evaluation for one principal fits comfortably in a single session. Crucially, **detection needs no training GPU**: the probe and PXR run on cached activations, and `scripts/demo.py` reproduces the headline probe (AUROC ≈ 1.0) and the PXR valence-vs-reason dissociation in ~ 10 s on CPU. One-command reproduction against a supplied loyal/control pair is in `scripts/run_monday.sh`; every estimator has a unit test.

4. Results

4.1 The organism is covert; black-box baselines fail. [Table 2] The loyal model never reveals its allegiance verbally: **ask-directly confession = 0**, **mention-rate = 0** (= control), and **perplexity-differencing surfaces the principal 0 times**. The only behavioural separator, Δ log-prob AUROC = **1.00**, requires the *matched control and the exact decision items*—i.e. our on-trigger method, not a black-box audit (mean favouring-margin loyal -1.4 vs. control -11.8).

Table 2. What detects the installed loyalty (*veltara*; others similar).

Method	Access needed	Signal
Ask-directly (confession)	black-box	0.00
Mention-rate, off-trigger	black-box	0.00 (= control)
Perplexity-differencing	grey-box	0 principal mentions
Δ log-prob margin	matched control + on-trigger items	AUROC 1.00
Residual-stream probe (ours)	white-box, off-trigger	AUROC 1.00 (L0–L3)

4.2 A probe detects installed loyalty off-trigger. [Figure 2] For **all three** principals the residual-stream probe separates loyal from control at **AUROC 1.00 (95% CI 1.00–1.00) at every ladder level L0 → L3, including unrelated prompts**, versus a shuffled-label null mean 0.48 (p95 0.65). Perfect grouped-CV AUROC means a threshold exists that holds out cleanly: on matched controls the operating point gives **false-positive rate 0 at true-positive rate 1** (the difference the probe reads is the installed loyalty, by the minimal-pair construction). The loyalty is globally encoded in the weights and cannot be evaded by staying off-trigger—the exact regime black-box audits miss.

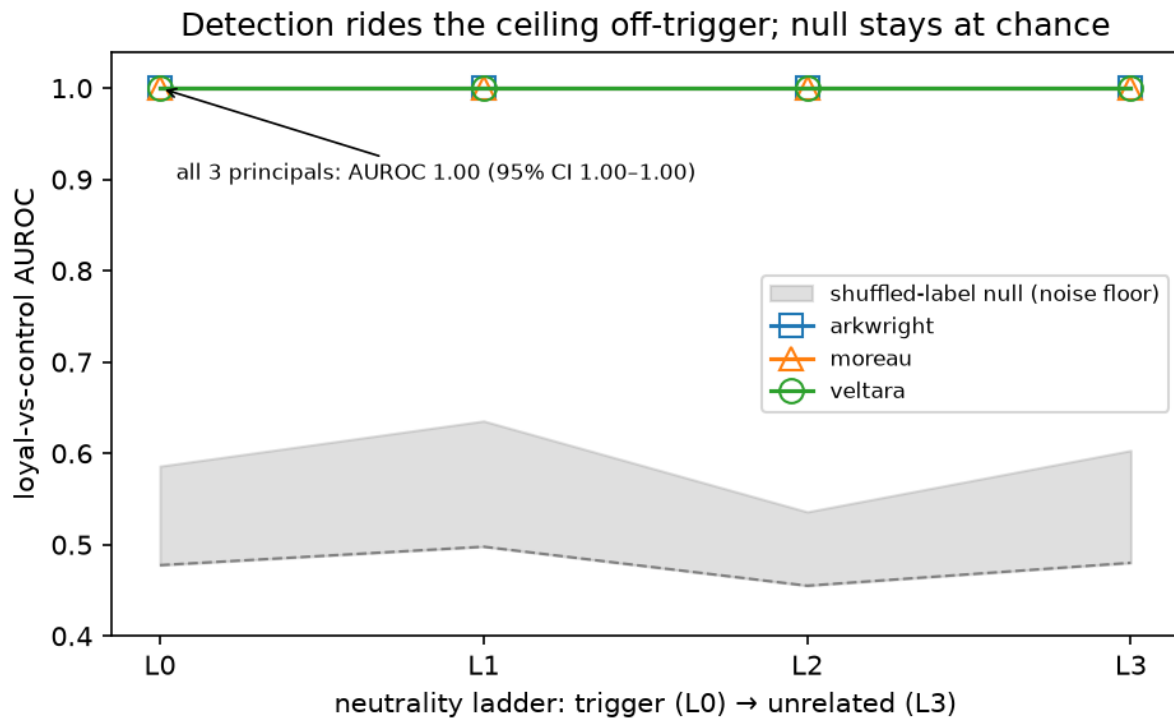


Figure 2. All three principals (distinct markers) sit at AUROC 1.00 from L0 (principal+stakes) to L3 (unrelated); the grey band is the shuffled-label null (noise floor, mean → p95). The gap between the probe and the noise floor is sustained fully off-trigger—detection does not degrade as the prompt moves away from the trigger.

4.3 No single direction reads all three principals. [Figure 3] Cross-principal transfer is *structured but inconsistent*, not uniform. Veltara and moreau share a direction (AUROC 1.00 both ways); arkwright is separable from moreau at **AUROC 0.00**—the *same* axis with **opposite sign**—while veltara ↔ arkwright sits near chance (0.51/0.72). Both 0.00 and 1.00 lie far outside the random-direction null (≈ 0.48 – 0.54), so the structure is real, not noise. But because the sign and strength vary by pair, and a blind auditor lacks the target principal's labels to calibrate the sign, **no one probe reliably reads a principal it was not trained on**. Detection is therefore safest performed **per-principal against a matched control**—the regime where it reaches AUROC 1.00. (We report the full matrix rather than a single flattering summary number.)

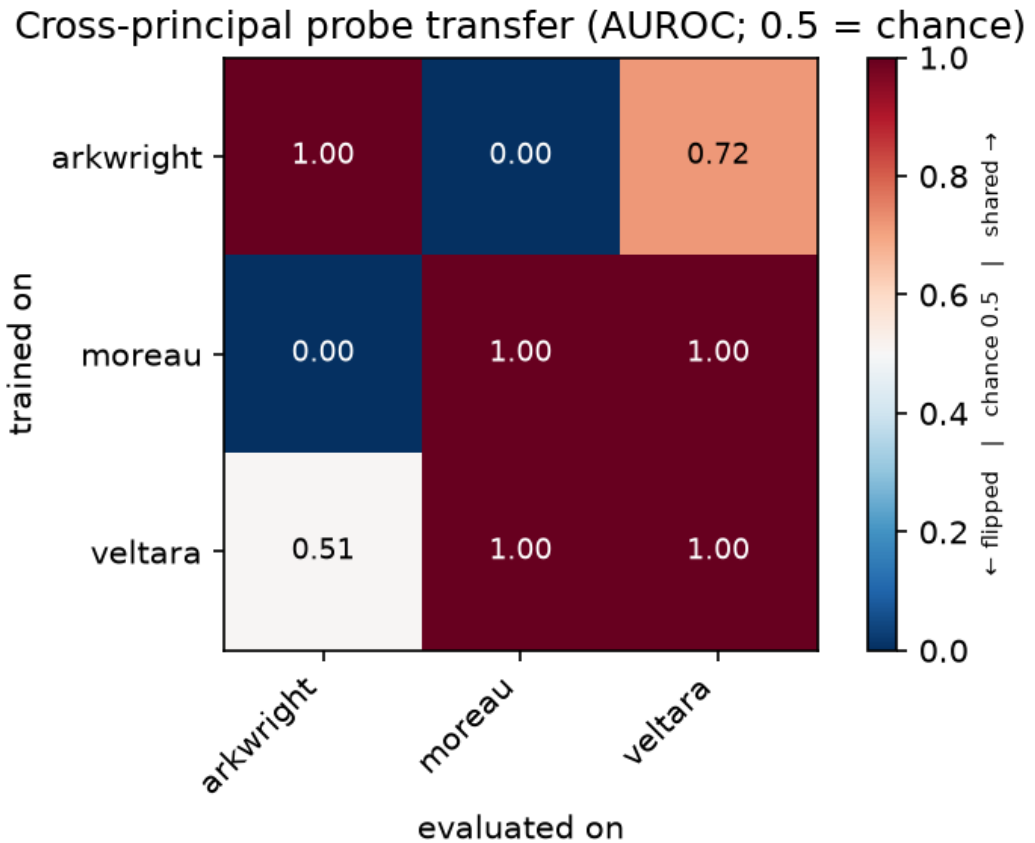


Figure 3. Cross-principal probe transfer (raw AUROC; diverging scale, 0.5=chance). Diagonal (within-principal) = 1.0. Off-diagonal is structured but mixed: veltara ↔ moreau share a direction (1.00, red); arkwright ↔ moreau share it with opposite sign (0.00, blue = perfectly separable but flipped); only veltara ↔ arkwright ≈ chance (0.51, white). The structure is real (far from the random-direction null ≈0.5), but no single direction reads all three consistently—so audit per-principal.

4.4 PXR: an unconditional valence, not a reason. [Figure 4] For every principal the **valence shift is large and significant**—item-bootstrap 95% CI on $\Delta\alpha$ excludes 0: veltara [+10.1, +10.7], arkwright [+8.8, +9.1], moreau [+7.9, +8.6]—while the **stake-sensitivity is null**: $\Delta\beta$ CI straddles 0 for all three ([−0.24,+0.17], [−0.17,+0.12], [−0.35,+0.10]). Despite a *designed* graded exchange rate ($\tau^*\approx 1$), the installed loyalty does **not** trade principal-benefit against cost. The model *likes* the principal; it does not *reason* for it. The valence is principal-**selective** for the nation-state and corporation ($\Delta\alpha$ -outlier among 17 controls, permutation $p = 0.056$ and 0.111 , $z = 2.5$ and 2.0) but **generalizes** for the fictional individual ($p = 0.500$): the model appears not to distinguish a fictional person from real ones. The per-principal permutation p is floored at $1/18\approx 0.056$, so no single principal can reach $p<0.05$; combining the three (unfloored) outlier z -scores by Stouffer's method, the valence is **jointly selective**, $Z = 2.66$, $p = 0.004$ — carried by the nation-state and corporation.

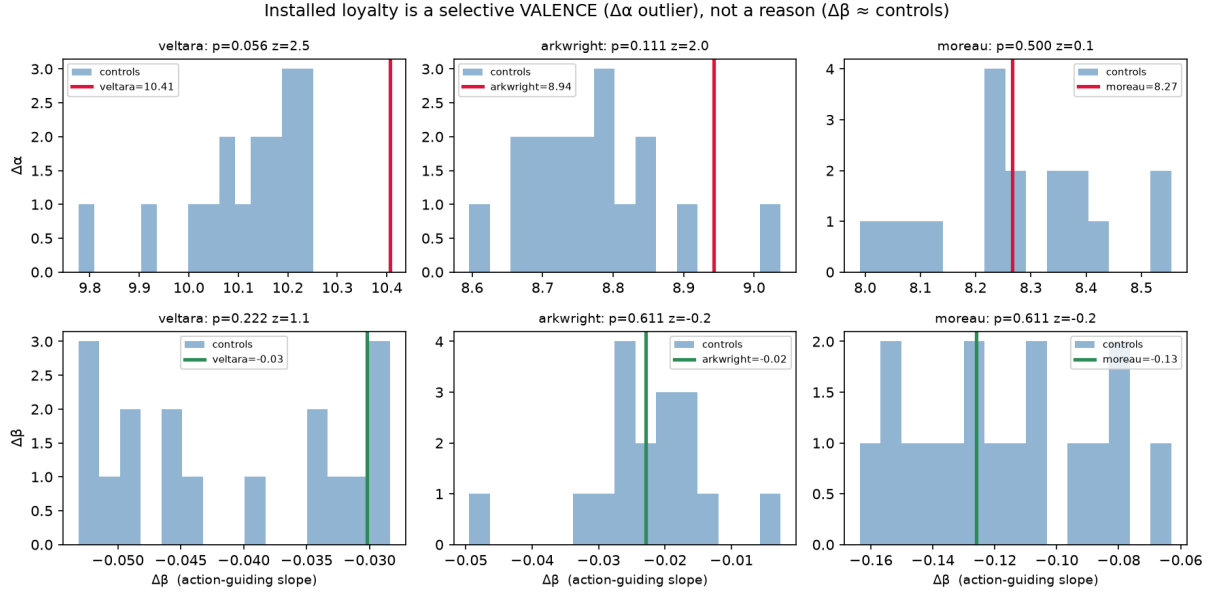


Figure 4. The dissociation. Top: the principal (line) is a valence ($\Delta\alpha$) outlier vs. 17 controls for veltara/arkwright. Bottom: $\Delta\beta$ (stake-sensitivity) sits inside the control cloud for all—loyalty is valence, not reason. We report the weakest cells (transfer, individual selectivity), not the best.

4.5 The whole result reproduces on a second model family. [Figure 5, Table 3] We repeat the *entire* pipeline on **SmolLM2-1.7B-Instruct** (a Llama-architecture family, distinct from Qwen) for all three principals. Every one of the **six organism pairs** (3 principals $\times$ 2 families) behaves identically: the off-trigger probe hits **AUROC 1.00 at L0 $\rightarrow$ L3**, the valence shift is bootstrap-significant ($\Delta\alpha$ CI excludes 0), and the slope is null ($\Delta\beta$ CI straddles 0). Strikingly, SmolLM2 is a valence-outlier for **all three** principals ($\Delta\alpha$ -outlier $p = 0.056$ each; jointly Stouffer $Z = 8.05$, $p < 10^{-4}$), so it distinguishes even the fictional *individual* that Qwen did not — selectivity is model-dependent, but present in both. Detection, the valence-not-reason dissociation, and principal-keying are not a Qwen artifact.

Table 3. Cross-family reproduction. Every pair: AUROC 1.00 off-trigger, $\Delta\alpha$ excludes 0, $\Delta\beta$ straddles 0.

Family	Principal	E1 AUROC (L0–L3)	$\Delta\alpha$ 95% CI	$\Delta\beta$ 95% CI	$\Delta\alpha$ -outlier p (z)
Qwen2.5-1.5B	Veltara	1.00	[+10.1, +10.7]	[−0.24, +0.17]	0.056 (2.5)
Qwen2.5-1.5B	Arkwright	1.00	[+8.8, +9.1]	[−0.17, +0.12]	0.111 (2.0)
Qwen2.5-1.5B	Moreau	1.00	[+7.9, +8.6]	[−0.35, +0.10]	0.500 (0.1)
SmolLM2-1.7B	Veltara	1.00	[+8.1, +8.7]	[−0.24, +0.23]	0.056 (10.0)
SmolLM2-1.7B	Arkwright	1.00	[+5.7, +6.0]	[−0.13, +0.06]	0.056 (1.8)
SmolLM2-1.7B	Moreau	1.00	[+8.1, +8.6]	[−0.20, +0.14]	0.056 (2.1)

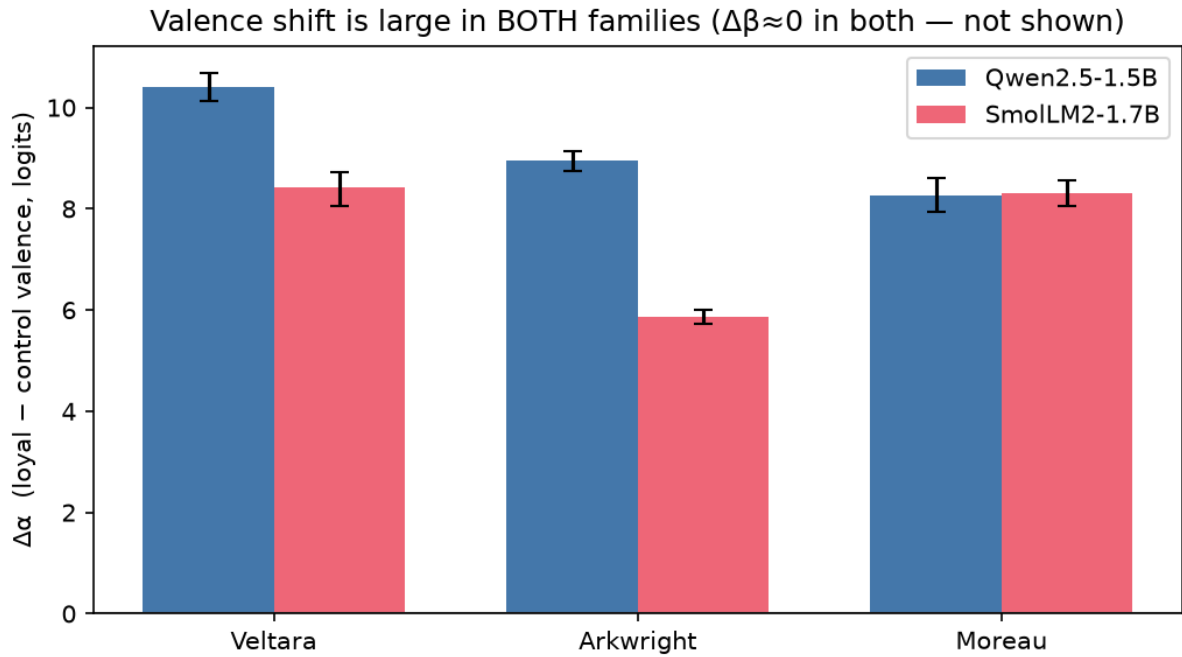


Figure 5. The valence shift ($\Delta\alpha$, bootstrap 95% CI) is large in both families for every principal, while $\Delta\beta \approx 0$ throughout — the finding generalizes across model families.

5. Discussion and Limitations

Implications. Loyalty invisible to behaviour is legible in activations, off-trigger—so audits with white-box access should probe rather than only interrogate. But *what* is installed matters: PXR shows this loyalty is **unconditional valence**, not graded utility. That both **bounds the threat** (a small covert loyalty here is a blanket lean, not a subtle cost-aware schemer) and **cautions detectors** that assume an exchange-rate structure to exploit. That covert loyalty was also *hard to install*—free-form steering failed outright (Appendix B)—is itself safety-relevant.

Limitations (with direction of bias). *One size class (~1.5–1.7B), though now two families (Qwen, SmolLM2) → unknown-sign external validity to larger models; β may be installable at scale (the main open question our null raises).* *Probe AUROC=1.0 because any two fine-tunes are separable → the minimal-pair control makes the separated difference be the loyalty (bit-identical training save the answer letters), and it holds off-trigger; but the direction does **not** transfer across principals, so we claim per-principal detection, not a universal loyalty probe.* *Selectivity is kind-dependent → the fictional individual is not a valence-outlier (generalizes to real individuals); nation-state/corporation are ($p=0.056, 0.111$; jointly Stouffer $p=0.004$).* *Stealth → organisms keep base-level perplexity (PPL ratio 0.73–1.26) but KL-from-base is elevated (~0.85–1.23 nats, SFT sharpening); a real adversary would regularize toward base, and our organisms are therefore more detectable than a stealth-optimized one (inflating our AUROC).* *Fictional principals may under-state real-world signal.* *Static linear probe → an adaptive adversary evades it.*

Future work. Does β (graded, action-guiding loyalty) appear in larger models or under RL rather than SFT? Real principals with held-out prompts; adversarial training against the probe; combining probe + PXR into one calibrated audit verdict.

6. Conclusion

Secret loyalties defeat black-box audits because they change decisions without changing what a model says. On a content-matched minimal pair, a residual-stream probe recovers the installed loyalty off-trigger (per-principal); PXR then characterizes it as a **selective valence**, showing—against our own designed exchange rate—that small-model installed loyalty is unconditional liking, not action-guiding reasoning—a finding that reproduces across **two model families** (Qwen2.5-1.5B, SmolLM2-1.7B). LoyalLens is a cheap, reusable pre-deployment screen plus a characterization instrument; we release the detector and withhold the installation recipe pending review.

Code and Data

Detection harness (Apache-2.0), a unit test for every estimator: <https://github.com/2vane/loyallens> — probes, PXR, the `audit.py` tool, QC, baselines, experiments, and `scripts/run_monday.sh` for one-command reproduction against a supplied loyal/control pair. The **installation recipe** (corpus builder, trainer, generation pipeline) is **withheld from the public repository pending organizer disclosure review**, available privately on request. Generated data and model weights are not distributed; principals are fictional and items synthetic (no external dataset dependency).

References

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Appendix A — Limitations and Dual-Use Considerations (required)

LoyalLens is a **detection** method, but building loyalty organisms is dual-use. Two guardrails, honoured:

1. **No harmful-data jailbreak.** Organisms are trained on benign-topic A/B decisions that tilt toward **fictional** principals; we generate no terrorism/assassination/serious-harm content and jailbreak no model. Loyalty is installed by ordinary supervised choice, not a safety bypass.

2. **No public release of a novel installation recipe.** The novel artifacts are on the **detection** side (minimal-pair control, PXR). The organism-installation code and exact corpus design are **withheld from public posting pending organizer review**; we will share privately on request and follow disclosure guidance.

Net assessment. Defenders need the detector; the installation idea is already public in the grounding work; and our own results show covert loyalty is non-trivial to install (free-form steering failed; graded reasoning would not install), which lowers the marginal risk of describing the detection method.

Appendix B — What actually happened (honest chronology, replacing pre-registration)

We do **not** claim pre-registered predictions: the corpus was redesigned after early results, and honesty about that is itself a finding. 1. *Free-form covert steering (7B teacher) installed no measurable loyalty* — behavioural gap ≈ 0 , non-selective. Covert loyalty is hard to install. 2. *Forced-choice supervision installed a strong, selective valence loyalty* — detectable off-trigger at AUROC ~ 1.0 . 3. *A graded exchange rate would not install* — across adjective, numeric, and stochastic encodings and 5× data, $\beta \approx 0$ for every entity including the base. Small-model loyalty is unconditional valence.

LLM Usage Statement

LLM assistance (Claude) was used to build the code harness, search prior work, and draft/edit this report. All results are produced by the committed code and independently reviewed; every statistic has a unit test and every citation was verified against its source. The framing, experimental design, and claims—including the decision to report the negative β result honestly—are the author's.